1. NAME OF THE MEDICINAL PRODUCT

LEVOPHED® (Norepinephrine Bitartrate) Injection, USP

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

LEVOPHED® (norepinephrine bitartrate) injection, USP, contains the equivalent of 1 mg base of Levophed® per 1 mL (4 mg/4 mL).

The drug product is comprised of a colorless or practically colorless liquid. The solution is colorless after dilution and not advised to be used if its color is pinkish or darker than slightly yellow or if it contains a precipitate.

3. PHARMACEUTICAL FORM

Solution for injection.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

For blood pressure control in certain acute hypotensive states (e.g., pheochromocytomectomy, sympathectomy, poliomyelitis, spinal anesthesia, myocardial infarction, septicemia, blood transfusion, and drug reactions).

As an adjunct in the treatment of cardiac arrest and profound hypotension.

4.2 Posology and method of administration

Norepinephrine Bitartrate Injection is a concentrated, potent drug which must be diluted in dextrose containing solutions prior to infusion. An infusion of LEVOPHED® should be given into a large vein (see Section **4.4 Special warnings and precautions for use**).

Restoration of Blood Pressure in Acute Hypotensive States

Blood volume depletion should always be corrected as fully as possible before any vasopressor is administered. When, as an emergency measure, intra-aortic pressures must be maintained to prevent cerebral or coronary artery ischemia, LEVOPHED® can be administered before and concurrently with blood volume replacement.

Diluent: LEVOPHED® should be diluted in 5 percent dextrose injection or 5 percent dextrose and sodium chloride injections. These dextrose containing fluids are protection against significant loss of potency due to oxidation. **Administration in saline solution alone is not recommended.** Whole blood or plasma, if indicated to increase blood volume, should be administered separately (for example, by use of a Y-tube and individual containers if given simultaneously).

Average Dosage: Add the content of the vial (4 mg/4 mL) of LEVOPHED® to 1,000 mL of a 5 percent dextrose containing solution. Each mL of this dilution contains 4 mcg of the base of LEVOPHED®. Give this solution by intravenous infusion. Insert a plastic intravenous catheter through a suitable bore needle well advanced centrally into the vein and securely fixed with adhesive tape, avoiding, if possible, a catheter tie-in technique as this promotes stasis. An IV drip chamber or other suitable metering device is essential to permit an accurate estimation of the rate of flow in drops per minute. After observing the response to an initial dose of 2 mL to 3 mL (from 8 mcg to 12 mcg of base) per minute, adjust the rate of flow to establish and maintain a low normal blood pressure (usually 80 mm Hg to 100 mm Hg systolic) sufficient to maintain the circulation to vital organs. In previously hypertensive patients, it is recommended that the blood pressure should be raised no higher than 40 mm Hg below the preexisting systolic pressure. The average maintenance dose ranges from 0.5 mL to 1 mL per minute (from 2 mcg to 4 mcg of base).

High Dosage: Great individual variation occurs in the dose required to attain and maintain an adequate blood pressure. In all cases, dosage of LEVOPHED® should be titrated according to the response of the patient. Occasionally much larger or even enormous daily doses (as high as 68 mg base or 17 vials) may be necessary if the patient remains hypotensive, but occult blood volume depletion should always be suspected and corrected when present. Central venous pressure monitoring is usually helpful in detecting and treating this situation.

Fluid Intake: The degree of dilution depends on clinical fluid volume requirements. If large volumes of fluid (dextrose) are needed at a flow rate that would involve an excessive dose of the pressor agent per unit of time, a solution more dilute than 4 mcg per mL should be used. On the other hand, when large volumes of fluid are clinically undesirable, a concentration greater than 4 mcg per mL may be necessary.

Duration of Therapy: The infusion should be continued until adequate blood pressure and tissue perfusion are maintained without therapy. Infusions of LEVOPHED® should be reduced gradually, avoiding abrupt withdrawal. In some of the reported cases of vascular collapse due to acute myocardial infarction, treatment was required for up to six days.

Adjunctive Treatment in Cardiac Arrest

Infusions of LEVOPHED® are usually administered intravenously during cardiac resuscitation to restore and maintain an adequate blood pressure after an effective heartbeat and ventilation have been established by other means. [LEVOPHED®'s powerful beta-adrenergic stimulating action is also thought to increase the strength and effectiveness of systolic contractions once they occur.]

Average Dosage: To maintain systemic blood pressure during the management of cardiac arrest, LEVOPHED® is used in the same manner as described under Restoration of Blood Pressure in Acute Hypotensive States.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to use, whenever solution and container permit.

Do not use the solution if its color is pinkish or darker than slightly yellow or if it contains a precipitate.

Avoid contact with iron salts, alkalis, or oxidizing agents.

4.3 Contraindications

- Hypersensitivity to norepinephrine (noradrenaline) or any of the excipients.
- Halothane anesthetics increase cardiac autonomic irritability and therefore seem to sensitize the myocardium to the action of intravenously administered epinephrine or noradrenaline. Hence, the use of LEVOPHED® during halothane anesthesia is generally considered contraindicated because of the risk of producing ventricular tachycardia or fibrillation.

4.4 Special warnings and precautions for use

Monoamine Oxidase Inhibitors and Tricyclic Antidepressants

Noradrenaline should be used with caution in patients receiving monoamine oxidase (MAO) inhibitors or tricyclic antidepressants, because severe, prolonged hypertension may result.

General

Particular caution should be observed in patients with coronary, mesenteric or peripheral vascular thrombosis because noradrenaline may increase the ischemia and extend the area of infarction. Similar caution should be observed in patients with hypotension following myocardial infarction, in patients with Prinzmetal's variant angina, and in patients with diabetes, hypertension or hyperthyroidism.

Noradrenaline should only be administered by healthcare professionals who are familiar with its use.

Noradrenaline should be used with caution in patients who exhibit profound hypoxia or hypercarbia.

Extravasation

Extravasation of the solution may cause local tissue necrosis. The infusion site should be checked frequently. If extravasation occurs, the infusion should be stopped, and the area should be infiltrated with phentolamine without delay.

Fluid Replacement

Noradrenaline should be used only in conjunction with appropriate blood volume replacement.

When infusing noradrenaline, the blood pressure and rate of flow should be checked frequently to avoid hypertension.

Prolonged administration of any potent vasopressor may result in plasma volume depletion which should be continuously corrected by appropriate fluid and electrolyte replacement therapy. If plasma volumes are not corrected, hypotension may recur when the infusion is discontinued, or blood pressure may be maintained at the risk of severe peripheral and visceral vasoconstriction (e.g., decreased renal perfusion) with reduced blood flow and tissue perfusion with subsequent tissue hypoxia and lactic acidosis and possible ischemic injury.

Withdrawal of Therapy

The noradrenaline infusion should be gradually decreased since abrupt withdrawal can result in acute hypotension.

Pediatric Use

The use of noradrenaline in children is not recommended.

Excipients

Sodium metabisulphite

This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

4.5 Interaction with other medicinal products and other forms of interaction

Influence on the Cardiovascular System

The use of noradrenaline with volatile halogenated anesthetic agents, monoamine oxidase inhibitors, tricyclic antidepressants, linezolid, adrenergic-serotoninergic drugs, or any other cardiac sensitizing agents is not recommended because severe, prolonged hypertension and possible arrhythmias may result.

Guanethidine

The effects of noradrenaline may be enhanced by guanethidine.

4.6 Fertility, pregnancy and lactation

Use in pregnancy

Noradrenaline may impair placental perfusion and induce fetal bradycardia. It may also exert a contractile effect on the pregnant uterus and lead to fetal asphyxia in late pregnancy. These possible risks to the fetus should therefore be weighed against the potential benefit to the mother.

Use in lactation

No information is available on the use of noradrenaline in lactation.

4.7 Effects on ability to drive and use machines

The effect of the medicinal product on the ability to drive or use machines has not been systematically evaluated. Patients should refrain from driving or using machines until they know that the medicinal product does not negatively affect these abilities.

4.8 Adverse effects (undesirable effects)

The adverse effects are reported within each system organ class (SOC).

Table 1. Adverse Drug Reaction Table

System Organ Class	Adverse Drug Reactions
Psychiatric disorders	Anxiety
Nervous system disorders	Headache
Cardiac disorders	Cardiogenic shock, arrhythmia*, bradycardia, stress cardiomyopathy[a]
Vascular disorders	Peripheral ischemia, gangrene, hypertension, plasma depletion
Respiratory, thoracic and mediastinal disorders	Dyspnea
General disorders and administration site conditions	Injection site extravasation, injection site necrosis

* When used in conjunction with cardiac sensitizing agents
[a] Frequency "not known"

4.9 Overdose

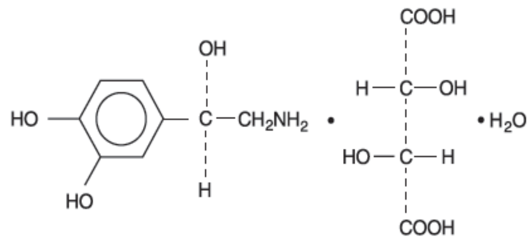
Overdose may result in severe hypertension, reflex bradycardia, marked increase in peripheral resistance, and decreased cardiac output. These may be accompanied by violent headache, pulmonary edema, photophobia, retrosternal pain, pallor, intense sweating and vomiting. In the event of overdose, treatment with noradrenaline should be withdrawn and appropriate corrective measures initiated.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Norepinephrine (sometimes referred to as *l-arterenol/Levarterenol* or *l-norepinephrine*) is a sympathomimetic amine which differs from epinephrine by the absence of a methyl group on the nitrogen atom.

LEVOPHED® is (-)-α-(aminomethyl)-3,4-dihydroxybenzyl alcohol tartrate (1:1) (salt) monohydrate (molecular weight 337.3 g/mol) and has the following structural formula:



LEVOPHED® is supplied in a sterile aqueous solution in the form of the bitartrate salt to be administered by intravenous infusion. Norepinephrine is sparingly soluble in water, very slightly soluble in alcohol and ether, and readily soluble in acids. Each mL contains 1 mg of norepinephrine base (equivalent 1.89 mg of norepinephrine bitartrate, anhydrous basis), sodium chloride for isotonicity, not more than 0.2 mg of sodium metabisulfite as an antioxidant. It has a pH of 3 to 4.5. The air in the vials has been displaced by nitrogen gas.

Mechanism of action

Norepinephrine is a peripheral vasoconstrictor (alpha-adrenergic action) and as an inotropic stimulator of the heart and dilator of coronary arteries (beta-adrenergic action).

Pharmacodynamics

The primary pharmacodynamic effects of norepinephrine are cardiac stimulation and vasoconstriction. Cardiac output is generally unaffected, although it can be decreased, and total peripheral resistance is also elevated. The elevation in resistance and pressure result in reflex vagal activity, which slows the heart rate and increases stroke volume. The elevation in vascular tone or resistance reduces blood flow to the major abdominal organs as well as to skeletal muscle. Coronary blood flow is substantially increased secondary to the indirect effects of alpha stimulation. After intravenous administration, a pressor response occurs rapidly and reaches steady state within 5 minutes. The pharmacologic actions of norepinephrine are terminated primarily by uptake and metabolism in sympathetic nerve endings. The pressor action stops within 1-2 minutes after the infusion is discontinued.

5.2 Pharmacokinetic properties

<u>Absorption</u>

Following initiation of intravenous infusion, the steady state plasma concentration is achieved in 5 minutes.

<u>Distribution</u>

Plasma protein binding of norepinephrine is approximately 25%. It is mainly bound to plasma albumin and to a smaller extent to prealbumin and alpha 1-acid glycoprotein. The volume of distribution is 8.8 L. Norepinephrine localizes mainly in sympathetic nervous tissue. It crosses the placenta but not the blood-brain barrier.

Elimination

The mean half-life of norepinephrine is approximately 2.4 min. The average metabolic clearance is 3.1 L/minutes.

Metabolism

Norepinephrine is metabolized in the liver and other tissues by a combination of reactions involving the enzymes catechol-O-methyltransferase (COMT) and MAO. The major metabolites are normetanephrine and 3-methoxyl-4-hydroxy mandelic acid (vanillylmandelic acid, VMA), both of which are inactive. Other inactive metabolites include 3-methoxy-4-hydroxyphenylglycol, 3,4-dihydroxymandelic acid, and 3,4-dihydroxyphenylglycol.

Excretion

Noradrenaline metabolites are excreted in urine primarily as sulphate conjugates and, to a lesser extent, as glucuronide conjugates. Only small quantities of norepinephrine are excreted unchanged.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Sodium Chloride
Sodium Metabisulfite
Water for Injection
Carbon Dioxide
Nitrogen

6.2 Incompatibilities

Infusion solutions containing noradrenaline (norepinephrine) acid tartrate monohydrate have been reported to be incompatible with iron salts, alkalis and oxidising agents, barbiturates, chlorpheniramine, chlorothiazide, nitrofurantoin, phenytoin, sodium bicarbonate, sodium iodide, streptomycin, sulfadiazine and sulfafurazole.

Incompatibility with insulin has also been reported.

6.3 Shelf-life

Please refer to carton.

6.4 Special precautions for storage

Store below 30°C

Protect from light.

6.5 Nature and contents of container

5 mL vials (4 mL fill, 4 mg/4 mL) in boxes of 10 (NDC No. 0409-3375-04).

Some product strengths or pack sizes may not be available in your country.

7. MANUFACTURER

Hospira Inc.,
1776 North Centennial Drive,
McPherson Kansas, 67460 United States

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